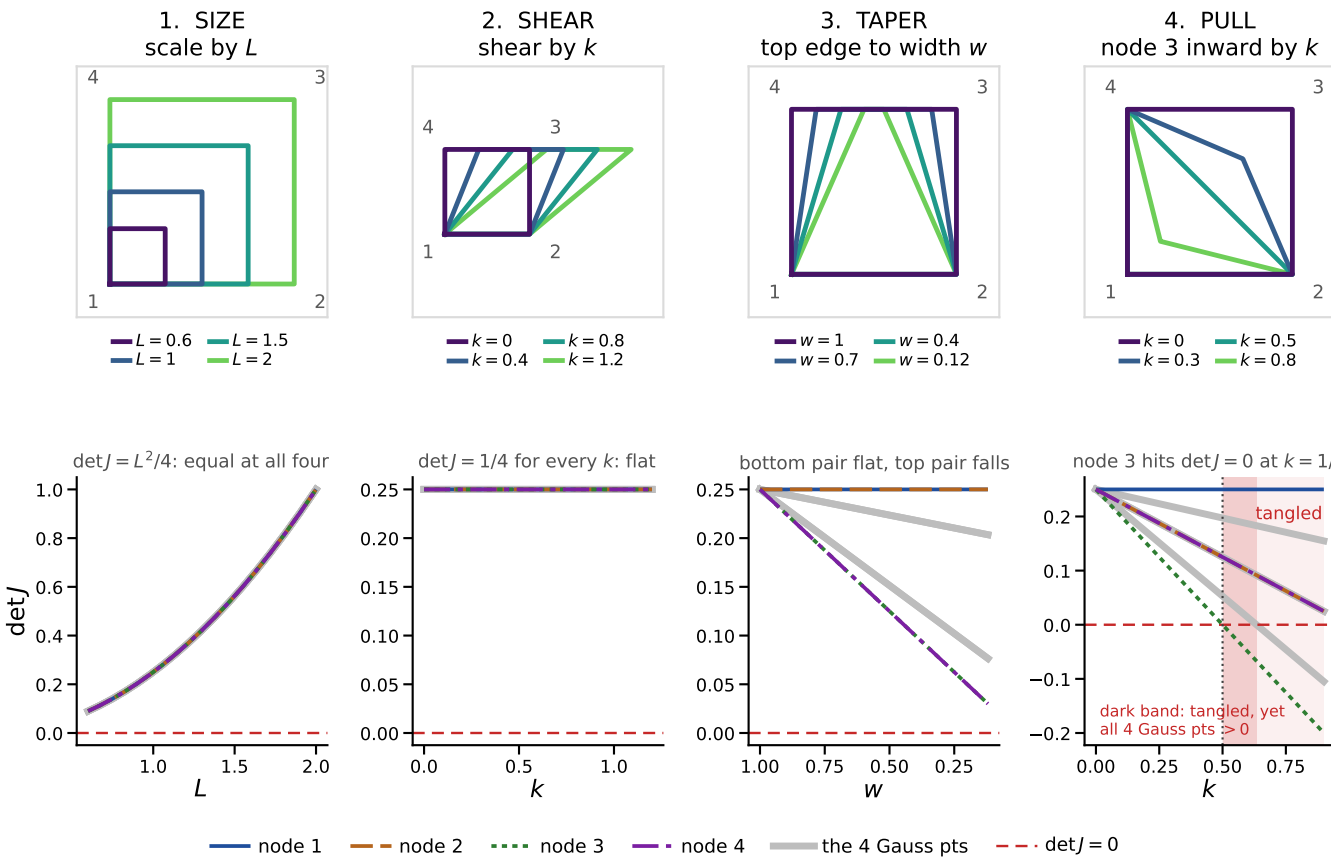
1. SIZE
scale by L
4
3
1
2
L = 0.6
L = 1
L = 1.5
L = 2
2. SHEAR
shear by k
4
3
1
2
k = 0
k = 0.4
k = 0.8
k = 1.2
3. TAPER
top edge to width w
4
3
1
2
w = 1
w = 0.7
w = 0.4
w = 0.12
4. PULL
node 3 inward by k
4
3
1
2
k = 0
k = 0.3
k = 0.5
k = 0.8
detJ = L²/4: equal at all four
detJ
L
detJ = 1/4 for every k: flat
k
bottom pair flat, top pair falls
w
node 3 hits detJ = 0 at k = 1/2
tangled
dark band: tangled, yet
all 4 Gauss pts > 0
k
node 1
node 2
node 3
node 4
the 4 Gauss pts
detJ = 0